MILITARY SPECIFICATION

RADIO SET AN/ARC-124(*)

This Amendment forms a part of Military Specification MIL-R-81299(WP) dated 1 August 1965 and has been approved by the Naval Air Systems Command, Department of the Navy.

By this amendment basic MIL-R-81299(WP) is changed to MIL-R-81299(AS), and throughout the specification Bureau of Naval Weapons shall be changed to Naval Air Systems Command

Paragraph 2.1, Under "SPECIFICATIONS": Delete "MIL-E-4682," "MIL-R-22256," and "MIL-R-23094" and their titles.

Paragraph 2.1, Above "STANDARDS": Add

"NASC

WR-97 Aircraft Electronics Equipment, Reliability Requirements for Design of"

Paragraph 2.1, Under "STANDARDS, Military": Add

"MIL-STD-781 Reliability Tests, Exponential Distribution"

Paragraph 3.1.1, After "materials": Add

"including tubes, transistors, and diodes."

Paragraph 3.1.2: Delete

Paragraph 3.2.1: Delete "Bureau of Naval Weapons, Code RAAV-331" and substitute

"Naval Air Systems Command."

Paragraph 3.2.3.3: Delete and substitute

"3.2.3.3 Reliability in Mean Time Between Failures (MTBF) - The equipment shall have 1000 hours of mean (operating) time between failures at a message rate of 10 message cycles per second (10 transmit messages alternated with 10 receive messages), when tested and accepted as outlined under the requirements of 4.4.3. The equipment shall be planned, designed, analyzed, and the reliability reported as outlined in WR-97."

FSC 5821

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Paragraph 3.2.16, last sentence: Delete and substitute

"Application of AC primary power to the equipment will be through a 28-volt DC relay, the ground side of which will be controlled by the associated digital communications system."

Paragraph 3.2.16: Add

"(2) DC Power 28 V, Category "B", 1 amp. max."

Paragraph 3.3.3.3, After "ohms": Add "±10%."

Paragraph 3.3.3.4: Delete "nominally". After "ohms" add

"±10%" and add the following sentence

"The input to the AN/ARC-124 will consist of binary marks and spaces wherein the transition from mark to space and vice versa will be accomplished between 50 and 80 microseconds measured at the 10% and 90% levels at the data input when the data source is terminated with 2500 ohms ±10%."

Paragraph 3.3.3.5, subparagraphs (a) and (b): After "ohms" add "non-inductive"

Paragraph 3.3.8: Delete "5" and substitute "10"

Paragraph 3.3.10: Delete "at least 10 volts" and substitute "10 volts, plus 2 minus 0 volts," and add the following:

The data DC reference level shift shall not exceed ±2.5 volts under all conditions of AN/ARC-124 frequency tolerances and environmental changes and under all conditions of control station frequency stability and tolerance the limits of which are ±2.5 KC."

Paragraph 3.3.13.4: Delete "5%" and substitute "10%"

Paragraph 3.3.13.5, line 2, After "deviation": Add "of ±20 kc"

Paragraph 3.3.18.1, After the second sentence add:

"The AN/ARC-124 shall accommodate input signal rise and fall times which are no slower than 2 microseconds as measured between the 10% and 90% amplitude points at the transceiver."

Paragraph 3.3.18.2: Delete "28" (two places) and substitute "34"

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Paragraph 3.3.18.4, second sentence: Delete and substitute

"The magnitude of the keying pulse shall be between 13 and 25 volts peak-to-peak and the maximum peak deviation of the voltage shall be within 0 volts and -30 volts."

After third sentence add:

"The signal will not exceed a duration of 11.2 milliseconds. The AN/ARC-124 shall accommodate input signal rise and fall times which are no slower than 2 microseconds as measured between the 10% and 90% amplitude points at the transceiver."

Paragraph 4.4.3, first sentence: Delete and substitute

"Reliability Assurance Tests shall be conducted using MIL-STD-781. Tests as required by both the Qualification Phase and the Sampling Phase shall be conducted."

Paragraphs 4.4.3.1, 4.4.3.2, and 4.4.3.3: Delete and substitute

"4.4.3.1 Qualification Phase - Prior to the acceptance of equipments under the contract or order, minimum of three (3) equipments shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests." The maximum number of equipments to be used shall be those listed in Table 5 of MIL-STD-781. For the Qualification Phase, Test Level F shall be used. The Accept-Reject Criteria for Test Plan I shall be used.

4.4.3.2 Reliability Sampling Phase Tests - Samples of the equipment shall be tested as outlined in MIL-STD-781, under the section entitled "Sampling Phase of Production Reliability Tests." For the Sampling Phase, Test Level F shall be used. The Accept-Reject Criteria for Test Plan IV shall be used to determine the length of the tests (until an accept or reject decision is reached).

4.4.3.2.1 Lot Size for Sampling Phase - The equipments constructed during one month shall be one lot.

4.4.3.3 Test Details - The test details such as the length of the Test cycle, the length of the heat portion of the cycle, special failure criteria, preventive maintenance to be allowed during the test etc., shall be part of the test procedures to be submitted and approved by the procuring activity prior to the beginning of the Qualification Test Phase of the Reliability Assurance Tests."

Paragraph 4.4.3.5: Delete

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